

Dataset Curation — Auto Routing Optimization Engine

Overview

The routing engine needs accurate, up-to-date information about drivers, loads, terminals, and delivery sites to build its daily dispatch plan. This document explains the origin of each piece of data, how it is refreshed, and how it is integrated into the system.

There are two source systems: CE Connect (our operational dispatch software) and a set of **Excel reference files** maintained on my local machine. Both feed into a cloud database (Supabase) that the routing engine reads from.

How Data Gets Into the System

A sync script runs automatically on my PC every 5 minutes. Each cycle it:

1. Reads the five Excel reference files from a local folder
2. Directly queries CE Connect's database for the latest loads and driver exceptions
3. Pushes all of it up to the cloud database (Supabase)

There is also a Force Sync button on the dashboard if you need to push an immediate update outside the 5-minute cycle.

The Six Datasets

1. Loads / Delivery Orders

Source: CE Connect database (direct SQL query — not Excel)

Refreshed: Every 5 minutes automatically

This is the main order feed. The sync script queries CE Connect directly and pulls every active load from the last 60 days. Each row represents one product on one delivery — multiple rows with the same CE ID are the same physical truck load with multiple products. The data includes the delivery date, customer name, site, terminal, delivery window times, assigned driver (if any), and current load status.

Loads already dispatched and in-progress (en route to rack, at the rack, en route to site, etc.) are preserved as-is and not overwritten once they're moving.

2. Driver Schedule

Source: Auto_Routing_Driver_Schedule.xlsx

Refreshed: This file is connected to CE Connect via ODBC, so it updates whenever the file is opened or refreshed in Excel. I refresh it before running dispatch each morning.

This file tells the engine who is working on a given day, what time their shift starts, which yard they're based out of (Dallas, Ft Worth, East Texas, etc.), and which board they belong to (TX-AM, TX-PM, FW-

AM, FW-PM, ET-AM). It also carries the `attendance_expected` flag — when a driver calls out or has an exception entered in CE Connect, the sync script picks that up separately and marks them as not working, without overwriting any manual attendance changes made in the dashboard.

3. Driver Terminal Cards

Source: `Driver_terminal_cards.xlsx`

Refreshed: Manually — updated when a driver gets a new terminal card or loses access.

Each row is a driver–terminal pairing, representing a fuel card that grants that driver access to load at that terminal. The engine uses this to determine which terminals a driver can pull from. If a driver has no card for a given terminal, they cannot be assigned a load that requires loading there.

4. Terminal Locations

Source: `terminal_locations.xlsx`

Refreshed: Manually — updated when a new terminal is added or location data changes.

This file contains every terminal in the network — name, address, GPS coordinates, and whether it is a diesel/wet terminal. The GPS coordinates are used by the engine to calculate drive times between the yard, terminal, and delivery sites. The `'is_diesel_wet'` column controls special sequencing rules (diesel-wet loads must be the first stop after the rack).

Note: Occasionally a terminal gets added directly in Supabase when it needs to be live immediately (e.g., Ft Worth Cargill, ODBC ID T-15-TX-0015). When that happens, the Excel file should be updated to match so they stay in sync.

5. Site Details

Source: `site_details.xlsx`

Refreshed: This file is connected to CE Connect via ODBC. I refresh it periodically, or when new customer sites come online.

Contains every delivery site — customer name, address, GPS coordinates, and whether the site requires a pump-certified driver. The engine uses GPS coordinates to calculate drive times from terminals to sites, and the pump certification flag to ensure only qualified drivers are matched to those deliveries.

There are currently 3,200 sites in this file.

6. Yard Locations

Source: `Yard_Locations.xls``

Refreshed: Rarely — only updated when a yard opens, closes, or moves.

Seven yard locations (Dallas, Ft Worth, East Texas, etc.) with GPS coordinates. The engine uses these as each driver's starting point when calculating their route for the day.

Driver Exceptions (Call-Outs / PTO)

Source: CE Connect database (separate query from loads)

Refreshed: Every 5 minutes automatically

When someone enters a driver exception in CE Connect, whether that's a call-out, PTO, or a day off, the sync script picks it up and marks that driver as not working (`attendance_expected = 0`) for that date. This happens after the driver schedules Excel sync runs, so exceptions always take priority. Manual attendance changes made directly in the dashboard are also preserved and not overwritten.

What the Routing Engine Actually Reads

When dispatch is run for a given date, the engine reads from Supabase and loads:

What	Table	Used for
Who's working today	driver_schedules	Active driver list, shift start times, board location
What needs to be delivered	load_details	All status = 1 unscheduled loads for that date
Which terminals each driver can use	driver_terminal_cards	Terminal eligibility check
Terminal GPS + diesel-wet flag	terminal_locations	Drive time calculations, sequencing
Site GPS + pump cert	site_details	Drive time calculations, driver-site matching
Driver's home base	yard_locations	Route start point

Loads already in progress (en route, at rack, at site) are seeded onto their assigned driver's route first, before the engine tries to assign any remaining unscheduled loads.

Where the Files Live

All Excel files are stored in a single folder on my PC. The path is set in the sync script's configuration file (.env). The sync script runs in the background as long as my PC is on and connected to the network.

[Local PC]

└─ Excel folder (ODBC-connected to CE Connect)

└─ Auto_Routing_Driver_Schedule.xlsx

└─ Driver_terminal_cards.xlsx

└─ terminal_locations.xlsx

└─ site_details.xlsx Overview

└─ Yard_Locations.xlsx

[CE Connect MySQL] —► sync script (every 5 min) —► [Supabase cloud DB] —► Routing Engine

[Excel files] —► sync script (every 5 min) —► [Supabase cloud DB]

Things That Require Manual Attention

- New terminals: Add to `terminal\_locations.xlsx` and note the ODBC terminal ID from CE Connect. If urgent, can be added directly to Supabase — but the Excel file must be updated too or it will be out of sync. - *we may be able to pull this now from the new CE import.*
- New sites: Should come through automatically via the ODBC-connected site_details Excel. If a site is missing GPS coordinates, the engine will still run but drive time for that site will be estimated at 0 miles — worth checking periodically. - *we may be able to pull this now from the new CE import*
- Driver terminal card changes: Update `Driver\_terminal\_cards.xlsx` manually when a driver's card access changes. - *confirmed this can be fed in from CE import. It will require an update to driver profiles in CE.*
- Driver restrictions: If a driver cannot deliver to a specific customer or site, that is managed directly in the dashboard (Restrictions tab) — it does not come from an Excel file. - *This can also be fed in from CE. I do not currently have this setup.*